

Simple subspaces expose weak evidence for Krylov transfer

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RESEARCH QUESTION

Does learning from Lanczos trajectories improve label efficiency for quantum ground states beyond credible simple baselines?

METHOD AND EVIDENCE BOUNDARY

Retained synthetic development study: 16 by 16 interior grid, one training seed, twelve in distribution test potentials, and 20 labeled examples.

A joint embedding model predicts trajectory representations; the potential encoder transfers to energy and wavefunction regression. Neural values come from retained aggregates; classical means are rechecked from per-example records.

WHY THIS MATTERS FOR PARSIMONY

A fixed free-box state and a nine-mode Rayleigh Ritz model reveal that high overlap can be easy on a smooth distribution. Physical residuals expose errors hidden by fidelity.

WHAT REMAINS UNRESOLVED

One training seed. Exploratory intervals exclude training seed uncertainty. The larger confirmatory experiment has not run.

High fidelity does not imply small physical error

Method	Fidelity	Residual
Krylov +20 labels	0.991650	40.190890
Scratch +20 labels	0.993945	29.043635
Fixed free box	0.994460	1.483805
Nine mode Ritz	0.999898	0.761666
Ridge +20 labels	0.999904	0.516766

Mean over twelve retained potentials. Residual: $||H \psi - E_{\text{true}} \psi|| / ||\psi||$, in energy units.

WHAT THE RESULT SUPPORTS

The retained Krylov model does not establish useful transfer. Its paired fidelity interval against free box includes zero; the nine-mode comparison favors the classical baseline within this corpus.

Ritz uses the query Hamiltonian and has a different inference cost. These comparisons do not establish equal compute dominance.